SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

The Role of Artificial Intelligence and Machine Learning in Cloud Platforms Discuss (AI/ML cloud services), and industry applications.

CSA1514 - CLOUD COMPUTING AND BIG DATA ANALYTICS

Guided By:
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AI/ML CLOUD SERVICES: THREE LAYERS



AI Services & APIs

Pre-built vision, speech, language and generative AI

Amazon Bedrock · Azure AI Services · Google Gemini API



ML Platforms

Managed build, train, deploy and MLOps

Amazon SageMaker · Azure Machine Learning · Google Vertex AI



AI Infrastructure

Elastic GPU/TPU compute, storage and networking

NVIDIA GPU instances · AWS Trainium · Google TPUs · Kubernetes

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CLOUD AI/ML ARCHITECTURE



1. Ingest

Batch and streaming data lands in data lakes



2. Prepare

Clean, label and store features centrally



3. Train

Distributed training on GPU/TPU clusters



4. Deploy

Serve models via APIs, containers or edge



5. Monitor

Track drift, accuracy and cost; retrain

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INDUSTRY APPLICATIONS

Healthcare

Medical imaging analysis and drug discovery

Finance

Real-time fraud detection and credit scoring

Retail

Personalised recommendations and demand forecasting

Manufacturing

Predictive maintenance and quality inspection

Automotive

Autonomous driving and connected-vehicle analytics

Customer Service

Chatbots and generative AI assistants

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BENEFITS, CHALLENGES

Benefits

- Elastic, on-demand scalability
- Pay-as-you-go cost model
- Faster time-to-market
- Built-in MLOps and security

Challenges

- High GPU and training costs
- Data privacy and compliance
- Vendor lock-in risk
- Skills gap and model governance

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THANK YOU

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